Do not write outside the box

**0 2** This question is about salts and electrolysis.

A student wants to make copper chloride crystals.

The student adds excess copper oxide to some hot acid.

The student stirs the mixture.

**0 2 . 1** Which acid should the student use?

**[1 mark]**

Tick (✓) **one** box.

Hydrochloric acid ☐

Nitric acid ☐

Sulfuric acid ☐

**0 2 . 2** Suggest how the student would know that excess copper oxide has been added.

**[1 mark]**

_____

_____

**Question 2 continues on the next page**

Do not write outside the box

**0 2 . 3** There are four more stages, **A**, **B**, **C** and **D**, to make copper chloride crystals.

The stages **A**, **B**, **C** and **D** are not in the correct order.

| | |
|---|---|
| Stage **A** | Partially evaporate by heating with a water bath |
| Stage **B** | Filter the mixture into an evaporating basin |
| Stage **C** | Leave to crystallise |
| Stage **D** | Remove and dry the crystals |

Put stages **A**, **B**, **C** and **D** in the correct order.

**[2 marks]**

First stage _____

Second stage _____

Third stage _____

Fourth stage _____

**0 2 . 4** Molten copper chloride can be electrolysed.

State the product at each electrode when molten copper chloride is electrolysed.

**[2 marks]**

Negative electrode _____

Positive electrode _____

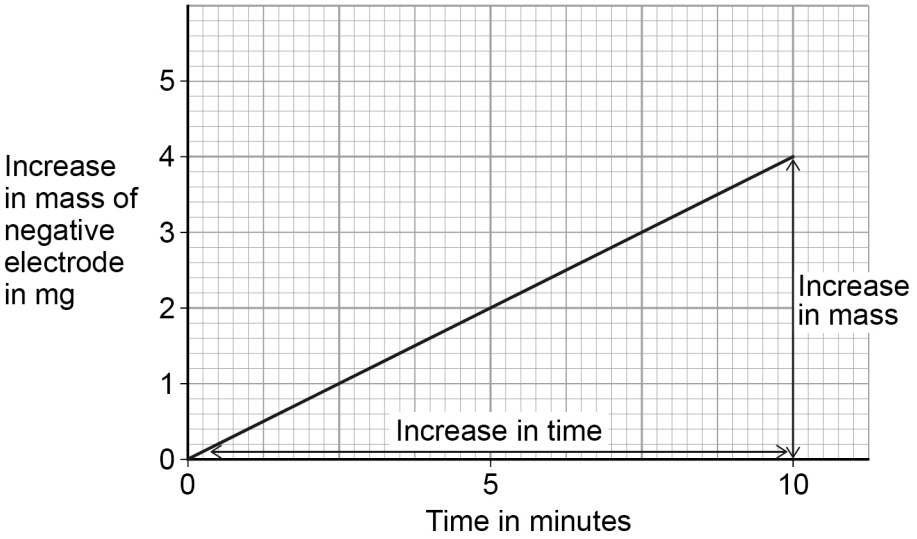
Do not write outside the box

**0 2 . 5** A solution of copper chloride is electrolysed.

**Figure 3** shows a graph of the increase in mass of the negative electrode.

This increase is shown over a time of 10 minutes.

**Figure 3**




Calculate the gradient of the line in **Figure 3**.

Use the equation:

$$\text{Gradient} = \frac{\text{increase in mass in mg}}{\text{increase in time in minutes}}$$

**[3 marks]**

Increase in mass _____

_____

Increase in time _____

_____

Gradient _____

_____

_____

_____

Gradient = _____ mg per minute

**Turn over ►**

Do not write outside the box

**0 2 . 6** Aluminium is produced by electrolysis of a molten mixture.

Complete the sentence.

Choose the answers from the box.

**[2 marks]**

| | | | | | |
|---|---|---|---|---|---|
| **carbon** | **chloride** | **cryolite** | **oxide** | **sulfate** | **water** |

The molten mixture contains _____ and

aluminium _____ .

11